

JWST Measurements of Neutral Hydrogen Fractions and Ionized Bubble Sizes at $z = 7 - 12$ Obtained with Ly α Damping Wing Absorptions in 27 Bright Continuum Galaxies

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ABSTRACT

We present volume-averaged neutral hydrogen fractions $x_{\text{H I}}$ and ionized bubble radii R_{b} measured with Ly α damping wing absorption of galaxies at the epoch of reionization. We combine JWST/NIRSpec spectra taken by CEERS, GO-1433, DDT-2750, and JADES programs, and obtain a sample containing 27 bright UV-continuum ($M_{\text{UV}} < -18.5$ mag) galaxies at $7 < z < 12$. We construct 4 composite spectra binned by redshift, and find the clear evolution of softening break towards high redshift at the rest-frame 1216 Å, suggesting the increase of Ly α damping wing absorption. We estimate Ly α damping wing absorption in the galaxy spectra with realistic templates including Ly α emission and circum-galactic medium absorptions. Assuming the standard inside-out reionization picture having an ionized bubble with radius R_{b} around a galaxy embedded in the intergalactic medium with $x_{\text{H I}}$, we obtain $x_{\text{H I}}$ (R_{b}) values generally increasing (decreasing) from $x_{\text{H I}} = 0.53^{+0.18}_{-0.47}$ to $0.92^{+0.08}_{-0.10}$ ($\log R_{\text{b}} = 1.67^{+0.14}_{-0.16}$ to $-0.69^{+0.89}_{-0.24}$ comoving Mpc) at redshift $7.12^{+0.06}_{-0.08}$ to $9.91^{+1.49}_{-1.15}$. The redshift evolution of $x_{\text{H I}}$ indicates a moderately late reionization history consistent with the one previously suggested from the electron scattering of cosmic microwave background and the evolution of UV luminosity function with an escape fraction $f_{\text{esc}} \sim 0.2$. Our R_{b} measurements suggest that bubble sizes could be up to a few dex larger than the cosmic average values estimated by analytic calculations for a given $x_{\text{H I}}$, while our R_{b} measurements are roughly comparable with the values for merged ionized bubbles around bright galaxies predicted by recent numerical simulations.

Keywords: Galaxy evolution (594), Galaxy formation (595), High-redshift galaxies (734), Reionization(1383)

1. INTRODUCTION

Cosmic reionization scenarios have been investigated by estimating the neutral hydrogen fraction $x_{\text{H I}}$ of the universe at different redshifts. Most powerful way to probe the neutral hydrogen at the epoch of reionization is to look at Ly α damping wing absorption. Numerous works have investigated $x_{\text{H I}}$ from Ly α damping wing absorption using the brightest sources such as quasars (QSOs) and gamma-ray burst (GRBs) at $z \sim 6 - 8$ (e.g.,

Mortlock et al. 2011; Bañados et al. 2018; Schroeder et al. 2013; Davies et al. 2018; Greig et al. 2019; Wang et al. 2020; Totani et al. 2006, 2014). With the bright rest-frame UV-continuum, we can detect Ly α damping wing absorption by the intergalactic medium (IGM). $x_{\text{H I}}$ value at different redshift have been estimated by fitting the damping wing absorption profile for different $x_{\text{H I}}$ values. However, the number density of QSOs and GRBs are expected to be very small at the higher redshift, inhibiting us from accessing the ionization state of IGM at $z \gtrsim 7$ (e.g., Wang et al. 2019; Matsuoka et al. 2023).

To overcome the problem of lacking the bright objects at the high redshift, we investigate the fainter but much more numerous galaxies as alternative target to infer $x_{\text{H I}}$ from Ly α damping wing absorptions. To directly observe the Ly α damping wing absorption in the spectrum of galaxies at $z > 7$, we need to detect their rest-frame UV continuum. Spectroscopically detecting the rest-frame UV continuum at such a high redshift is very challenging because it requires deep spectroscopic data in near infrared (NIR). However, recent observations by the James Webb Space Telescope (JWST; Gardner et al. 2023; Rigby et al. 2023; Rieke et al. 2023; Jakobsen et al. 2022; Ferruit et al. 2022; Böker et al. 2023) have shown that it is capable of deep NIR spectroscopic observations. As presented in recent works, low-resolution (i.e., $R \sim 100$) NIRSpec/PRISM spectroscopic data is especially suited for detecting faint rest-frame UV continuum (e.g., Curtis-Lake et al. 2023; Nakajima et al. 2023; Harikane et al. 2023; Arrabal Haro et al. 2023a,b; Hsiao et al. 2023; Williams et al. 2023; Roberts-Borsani et al. 2022).

Curtis-Lake et al. (2023) are the first to demonstrate the power of the PRISM data to estimate Ly α damping wing absorption in the UV continua of high redshift galaxies. Curtis-Lake et al. (2023) have demonstrated the potential of the NIRSpec PRISM data to estimate the value of $x_{\text{H I}}$. However, their constraint on $x_{\text{H I}}$ is not very strict because the redshift for the object is not constrained due to the lack of emission line detections. Recently, Hsiao et al. (2023) have incorporated Ly α damping wing absorption in the spectral fitting of spectroscopically confirmed $z = 10.17$ lensed galaxy and constrained $x_{\text{H I}} > 0.9$ with a one sigma significance. In our work, we also take advantage of the multiple PRISM spectroscopy data with the redshift confirmation obtained by multiple surveys to systematically infer the $x_{\text{H I}}$ values at various redshift above $z = 7$ to capture the redshift evolution of the damping wing profile and corresponding $x_{\text{H I}}$ values.

This paper is constructed as follows. In Section 2, we describe our galaxy sample and the reduction of corresponding spectroscopic data. We construct composite spectra binned by redshift and fit for the Ly α damping wing absorption seen in the composite spectra in Section 3. We discuss the implications of the $x_{\text{H I}}$ estimates on understanding cosmic reionization history and future prospect in Section 4. We summarize this paper in Section 5. In this paper, we use the Planck cosmological parameter sets of the TT + TE + EE + lowE + BAO + lensing result (Planck Collaboration et al. 2020a): $h = 0.6766$, $\Omega_m = 0.3103$, $\Omega_\Lambda = 0.6897$,

$\Omega_b h^2 = 0.02234$, and $Y_p = 0.248$. All magnitudes are in the AB system (Oke & Gunn 1983).

2. DATA AND SAMPLE

We use spectroscopic datasets obtained in multiple observation programs; the Cosmic Evolution Early Release Science (CEERS; ERS-1345. PI. S. Finkelstein; Finkelstein et al. 2023), GO-1433 (PI. D. Coe; Hsiao et al. 2023), DDT-2750 (PI. P. Arrabal Haro; Arrabal Haro et al. 2023b), and the JWST Advanced Deep Extragalactic Survey (JADES; GTO-1210; PI. N. Lützgendorf; Bunker et al. 2023). We only used the PRISM data ($R \sim 100$) that covers $0.6\text{--}5.3\ \mu\text{m}$. The total exposure time for the PRISM spectra from CEERS, GO-1433, DDT-2750, and JADES data are 0.86, 3.7, 5.1, and 9.3-27.7 hours, respectively. As presented in Nakajima et al. (2023) and Harikane et al. (2023), all the PRISM data from CEERS, GO-1433, and DDT-2750 we use are reduced with the JWST pipeline version 1.8.5 with the Calibration Reference Data System (CRDS) context file of `jwst1028.pmap` or `jwst1027.pmap` with some improvements on the flux calibration, noise estimate, and the composition. For the JADES data, we use the publicly available reduced data from JADES group¹. Please see Bunker et al. (2023) for the detailed reduction processes of JADES data. We then select the galaxies with systemic redshifts above $z > 7$ confirmed with emission lines. For the $z < 9$ galaxies in CEERS, we adopt the spectroscopic redshift reported by Nakajima et al. (2023) in which the authors determine spectroscopic redshift via fitting [O III] $\lambda\lambda 4959, 5007$ and H β triplet simultaneously. For the galaxies at $z > 9$ obtained from observational program beside JADES, we adopt the spectroscopic redshift reported by Harikane et al. (2023). In Harikane et al. (2023), the authors derive the spectroscopic redshift for $z > 11$ object via [O II] $\lambda 3727$ (and [Ne III] $\lambda 3869$) for Maisie’s galaxy (CEERS2.588). For MACS0647-JD, Harikane et al. (2023) use H γ to determine spectroscopic redshift. For JADES objects, we adopt redshift reported in the public catalog released by the JADES team. The determination of the spectroscopic redshift listed in the public JADES is described in Bunker et al. (2023). In Bunker et al. (2023), the authors use emission lines detected with a signal to noise ratio of $S/N > 5$ to determine the spectroscopic redshift. Although the uncertainty of the spectroscopic redshift is not given in the literature, Jones et al. (2023) report that the spectroscopic redshift determined from PRISM data has a median off-

¹ <https://archive.stsci.edu/hlsp/jades>

set of 0.00388 with a standard deviation of 0.00628. In Section 4.5, we discuss on the effect of the uncertainty in the redshift confirmation with low resolution PRIMS data to our analysis. Please see Nakajima et al. (2023), Harikane et al. (2023), and Bunker et al. (2023) for further details on the data reduction and spectroscopic redshift determination. We further narrow the sample by selecting galaxies with data available in wavelength corresponding to rest-frame 1200 – 1250 Å to investigate Ly α damping wing features. We obtain a total of 35 galaxies as our galaxy sample candidates. Our sample candidates consist of 35 galaxies from $z = 7.045$ to 11.40, with the redshift distribution shown in Figure 1. In Table 1, we summarize properties of the sample candidate galaxies. We apply additional sample selection by setting an absolute UV magnitude cut of $M_{\text{UV}} < -18.5$ to omit faint ($M_{\text{UV}} > -18.5$) galaxies distributed mainly at $z < 8$ in our sample candidates. We also discard three of the object which Nakajima et al. (2023) only report 5σ upper limit value for UV magnitude for Hubble Space Telescope (HST) observations given that these objects are not detected by HST and lack JWST/NIRCam data. After this magnitude cut selection, we have a final galaxy sample consisted by 27 galaxies from $7.045 < z < 11.40$. Out of all 27 galaxies, spectroscopic data of 18, 1, 3, and 5 galaxies are obtained in CEERS, GO-1433, DDT-2750, and JADES programs, respectively. We present the complete PRISM spectra of galaxies in our final sample in the left panels of Figure 2 in the order of the highest to lowest redshift galaxies. The black line and grey region in each panel represents the observed flux and the error of each galaxy spectra normalized by an arbitrary unit (a.u.), respectively. We also show the wavelengths of Ly α determined from the systemic redshifts by the red vertical dashed lines. Next to the spectra of each galaxy, we present the emission line detected in the PRISM spectra with the highest S/N for each galaxy in the right hand panels. The systemic redshift values and the name of the emission line shown for each galaxy are written in between the panels.

The spectra of the galaxy in our final sample show various features around rest-frame Ly α wavelength. We show two contrasting example in Figure 3. The 2D and 1D spectra of CEERS2.588 (00021842) with systemic redshift at 11.04 (7.980) is shown in the left (right) panel. The red dotted lines represent the systemic redshift determined by emission line. The spectra of CEERS2.588 shows redshifted, softened Ly α break relative to systemic redshift, while we see blueshifted flux with 4.2σ significance beyond the rest-frame 1216 Å for 00021842. In Section 3.4, we physically interpret the di-

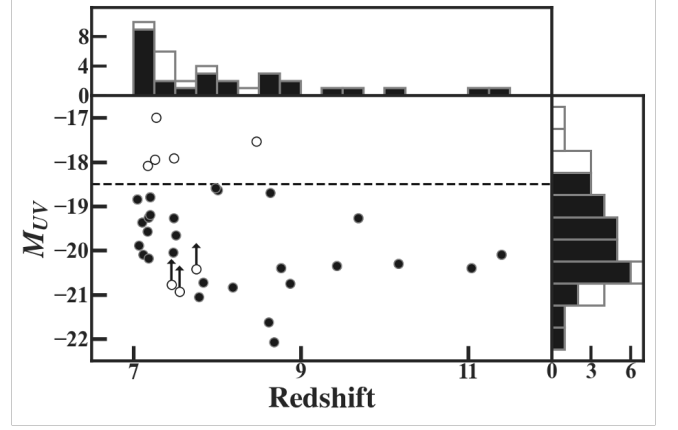


Figure 1. Redshift and UV-magnitude distribution of our galaxy sample. In the bottom panel, circles represent the redshift and UV-magnitude values for the galaxy observed by CEERS, GO-1433, DDT-2750, and JADES. The circle indicate the best-fit values while the upper circle with an upper arrow corresponds to one sigma upper lower-limit for the UV-magnitudes. The filled circle indicates the galaxy used in our analysis, while the open circle indicates the galaxies omitted from our sample after a selection based on the UV-magnitudes. The top (right) panel shows the number counts at each redshift (M_{UV}) bins for our galaxy sample. The fraction of bins represented by black represents the final sample after the absolute UV-magnitude selection.

verse spectral features around the rest-frame 1216 Å via spectral fitting.

3. METHODS

3.1. Ly α Damping Wing

To calculate the optical depth associated with Ly α damping wing absorption, we assume that galaxy resides at the center of the fully ionized bubble with radius R_b and the surrounding IGM with a constant x_{HI} until the end of EoR, z_n . We fix the redshift corresponding to the end of reionization at $z_n = 5$ in our calculations. Note that changing z_n by ± 1 have a negligible effect on IGM Ly α damping wing absorption profile.

To calculate the optical depth of Ly α damping wing at each observed wavelength λ_{obs} , $\tau(\lambda_{\text{obs}})$, we adopt the equation 2 of Totani et al. (2006) that is based on the formulation of Miralda-Escudé (1998):

$$\tau(\lambda_{\text{obs}}) = \frac{x_{\text{HI}} \Lambda_{\alpha} \lambda_{\alpha} \tau_0(z_s)}{4\pi^2 c} \left(\frac{1 + z_{\text{obs}}}{1 + z_s} \right)^{3/2} \times \left[I \left(\frac{1 + z_b}{1 + z_{\text{obs}}} \right) - I \left(\frac{1 + z_n}{1 + z_{\text{obs}}} \right) \right], \quad (1)$$

where Λ_{α} , λ_{α} , c , and τ_0 , are the decay constant for the Ly α resonance, the rest-frame Ly α wavelength, the

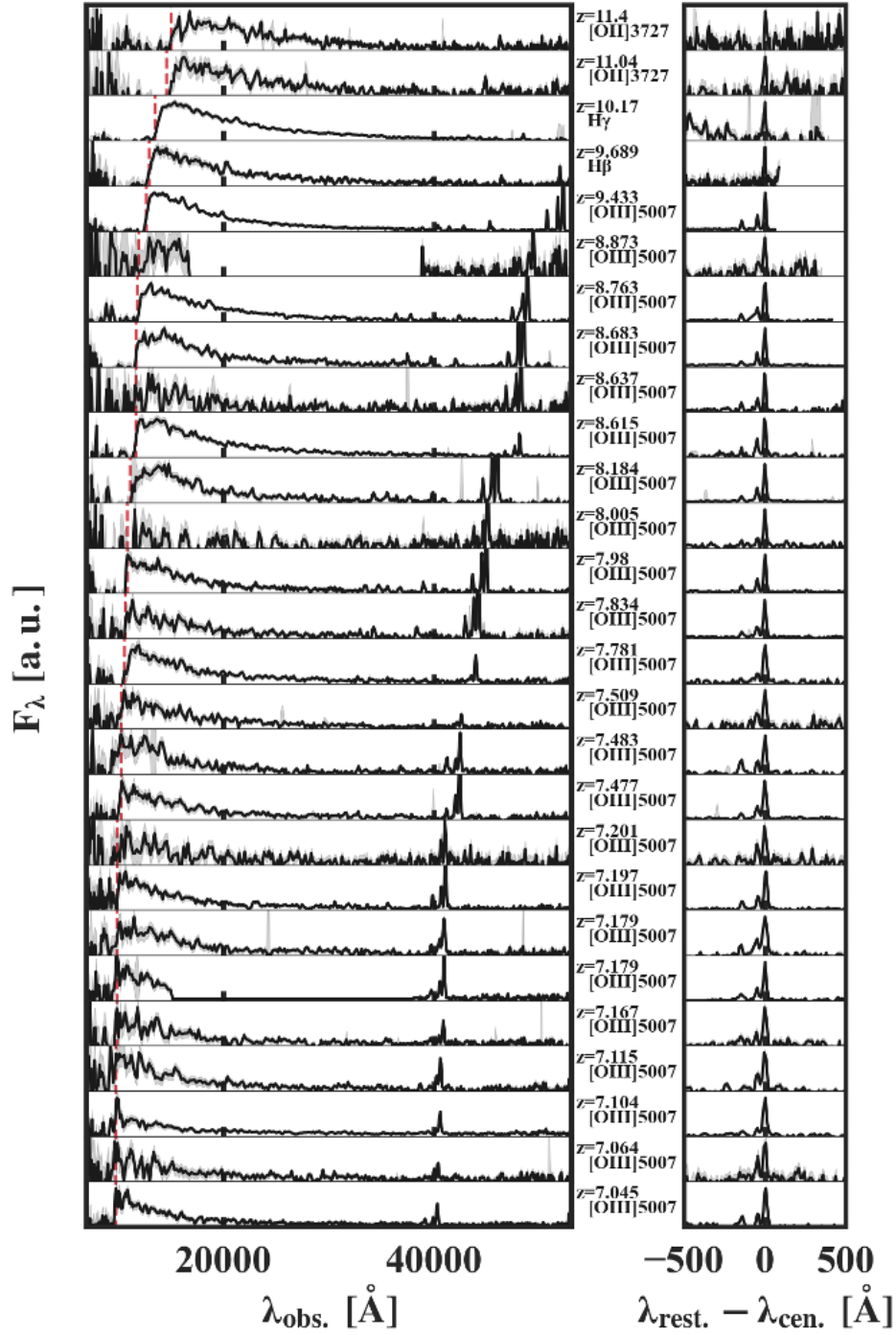


Figure 2. Full spectra of galaxy samples. (Left): The x-(y)-axis corresponds to the observed wavelength (arbitrary normalized flux density). The black lines and grey regions represent the fluxes of spectra and their uncertainties, respectively. The red dashed line represents the observed Ly α wavelength for each object. (Right) The highest S/N emission lines detected from each spectra in our sample. The type of emission line shown and the name of the corresponding galaxy is shown between the right and left panels. The x-axis represents the difference between the rest-frame wavelength $\lambda_{\text{rest.}}$ and the central wavelength $\lambda_{\text{cen.}}$ of the optical emission line.

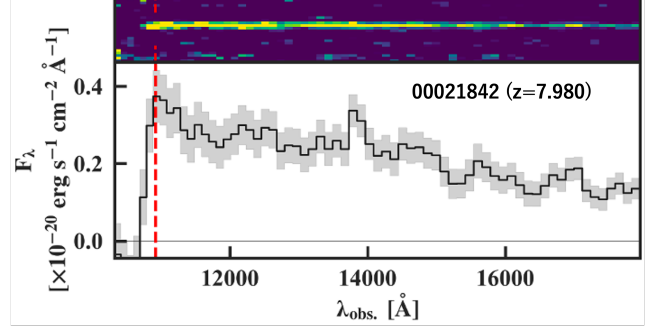
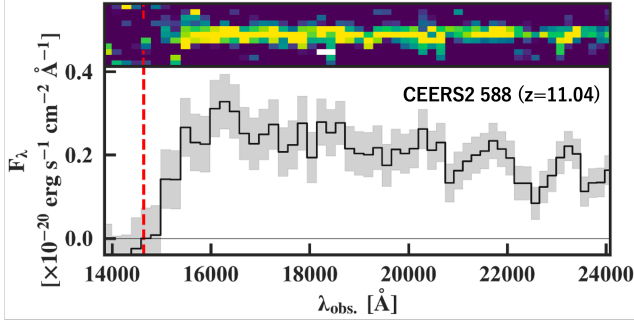


Figure 3. 2D (top) and 1D (bottom) spectra of CEERS2_588 (left) and 00021842 (right). The black line and grey region represent the observed flux and its uncertainty, respectively. The red dashed line represents the wavelength of Ly α at the systemic redshift determined by non-Ly α emission lines.

speed of light, and the Gunn-Peterson optical depth, respectively. z_b is defined as the redshift corresponding to the edge of the ionized bubble with radius R_b , whereas z_{obs} is a value defined as $\lambda_{\text{obs}}/\lambda_{\alpha} - 1$. $I(x)$ is an integration defined as follows:

$$I(x) = \frac{x^{9/2}}{1-x} + \frac{9}{7}x^{7/2} + \frac{9}{5}x^{5/2} + 3x^{3/2} + 9x^{1/2} - \frac{9}{2} \ln \frac{1+x^{1/2}}{1-x^{1/2}}. \quad (2)$$

We calculate τ_0 using the following formula also given by Miralda-Escudé (1998):

$$\tau_0(z_s) = \frac{3\lambda_{\alpha}^3 \Lambda_{\alpha} n_0}{8\pi H(z_s)}, \quad (3)$$

where n_0 and $H(z)$ are the hydrogen number density and the Hubble parameter as the function of redshift. We calculate n_0 using the critical density of the universe, baryon density parameter, and primordial helium abundance. We show how the absorption profile of the $z = 8$ changes with different x_{HI} and R_b values in Figure 4. As shown in Figure 4, increasing x_{HI} value extends the absorption feature to a redder side of 1216 Å while increasing R_b value shifts the absorption profile to the bluer wavelength. Notably, fluxes on the bluer side of the 1216 Å can be transmitted if the bubble size is sufficiently large (i.e., $R_b > 10$ comoving Mpc; cMpc).

3.2. Stacking Spectra

We construct composite spectra by stacking individual spectra to maximize the signal to noise ratio. We divide the galaxy sample into four subsamples by binning according to the redshifts so that each subsample has at least six galaxies. We label the subsamples as #1, #2, #3, and #4 bins in the order of the lower to higher redshift. The redshift range for #1, #2, #3, and #4 bins are 7.045-7.179, 7.197-7.781, 7.834-8.683, and 8.763-11.40, respectively. We define representative

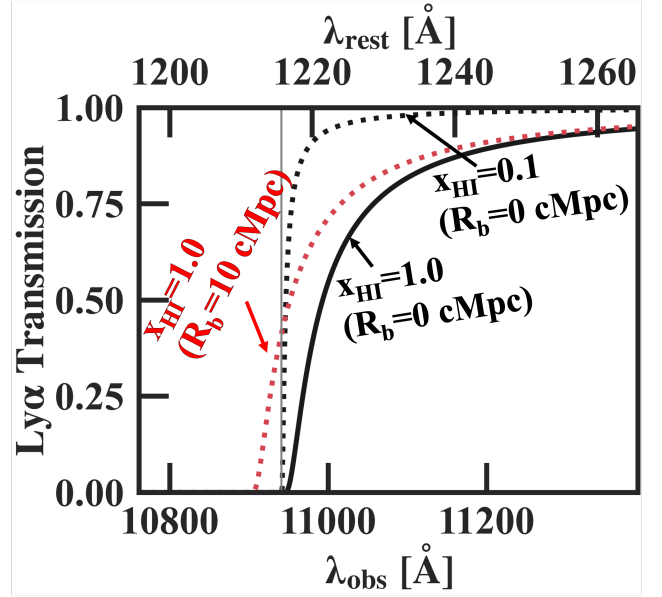


Figure 4. Example of Ly α damping wing profile for $z = 8$. The top and bottom axes are the rest frame and observed wavelengths, respectively. The y-axis corresponds to the transmission fraction of the flux at each wavelength. The solid (dotted) black line corresponds to the absorption profile with $x_{\text{HI}} = 1.0$ (0.1) and $R_b = 0$ cMpc. The red dotted line corresponds to the absorption profile with $x_{\text{HI}} = 1.0$ and $R_b = 10$ cMpc. The gray vertical line corresponds to the rest-frame Ly α wavelength.

redshift values $\langle z \rangle$ of each redshift bin by the average redshift value among the galaxies in each subsample.

For stacking individual spectra of each subsample, we have shifted the galaxy spectra into rest-frame wavelength based on the systemic redshift and resampled to a common wavelength pixels with a pixel size of 25 Å, a value considerably similar to that of the resolution of PRISM at $< 2 \mu\text{m}$ ($R \sim 30$), using linear interpolation. We have taken error weighted mean flux values at wavelength pixel to calculate the stacked spectra. We estimate the uncertainty of the stacked spectra via 10000

Table 1. Sample Properties

Name	R.A.	Decl.	z_{spec}	M_{UV}	Ref.
	(deg)	(deg)		(mag)	
(1)	(2)	(3)	(4)	(5)	(6)
00020961	53.13423	-27.76891	7.045	-18.840	l,m
CEERS_00542	214.831624	52.831505	7.064	-19.89	h
CEERS_00044	215.001115	53.011269	7.104	-19.36	h
CEERS_00534	214.859117	52.85364	7.115	-20.1	h
CEERS_00829	214.861594	52.876159	7.167	-19.57	h
CEERS_80374	214.898074	52.824895	7.174	-18.09	h
CEERS_00439	214.825364	52.863065	7.179	-19.25	h
CEERS_00498	214.813045	52.834249	7.179	-20.18	h
10013905	53.11833	-27.76901	7.197	-18.800	l,m
CEERS_01038	215.039697	52.9015971	7.201	-19.2	h
00008079	53.15283	-27.80194	7.260	-17.950	l,m
10013682	53.16746	-27.77201	7.275	-17.000	l,m
CEERS_01163	214.990468	52.9719902	7.455	> -20.78	h
CEERS_80432	214.812056	52.746747	7.477	-20.05	h
CEERS_80372	214.927798	52.850003	7.483	-19.27	h
CEERS_80239	214.896054	52.869853	7.487	-17.92	h
CEERS_80445	214.843115	52.747886	7.509	-19.66	h
CEERS_00689	214.999053	52.9419767	7.552	> -20.93	h
CEERS_00686	215.150862	52.9895618	7.752	> -20.42	h
CEERS_01023	215.188413	53.0336473	7.781	-21.06	h
CEERS_01027	214.882994	52.8404159	7.834	-20.73	h
00021842	53.15682	-27.76716	7.980	-18.590	l,m
CEERS_00003	215.005189	52.99658	8.005	-18.64	h
CEERS_01149	215.089714	52.9661828	8.184	-20.83	h
00008013	53.16446	-27.80218	8.473	-17.540	l,m
EGS_z10.44164	215.218762	53.0698619	8.615	-21.63	f,j
					h,d
CEERS_90671	214.961276	52.842364	8.637	-18.7	b,d
CEERS_1019	215.035391	52.8906618	8.683	-22.07	h,j,k
					i,g,d
CEERS_43833	214.938625	52.911750	8.763	-20.40	c,h
CEERS1_6059	215.011706	52.988303	8.873	-20.75	c,d,h
10058975	53.11243	-27.77461	9.433	-20.355	l,m
00006438	53.16735	-27.80750	9.689	-19.270	l,m
MACS0647-JD [†]	101.982213	70.2432667	10.17	-20.3	d,e
CEERS2_588 [‡]	214.906625	52.9115278	11.04	-20.4	d
Maisie's Galaxy [‡]	214.943167	52.9424417	11.40	-20.1	a,d
(CR2-z12-1)					

NOTE— (1): Name. (2): Right ascension of the object's coordinate. (3): Declination of the object's coordinate. (4): Spectroscopic redshift. (5): Absolute UV magnitude. (6): References for spectroscopic redshifts and UV magnitude: a.) Arrabal Haro et al. (2023a), b.) Arrabal Haro et al. (2023b), c.) Fujimoto et al. (2023a), d.) Harikane et al. (2023), e.) Hsiao et al. (2023), f.) Larson et al. (2022), g.) Larson et al. (2023), h.) Nakajima et al. (2023), i.) Sanders et al. (2023), j.) Tang et al. (2023), k.) Zitrin et al. (2015), l.) Bunker et al. (2023), m.) Jones et al. (2023)

All spectroscopic data are from CEERS program except for the spectroscopic data of the object name with [†] and [‡] are from GO-1433 and DDT-2750 programs, respectively.

iterations of bootstrap resampling. During each resampling, we fluctuate the observed flux at each pixel with Gaussian noise accordingly to the original error spectrum to consider the measurement error as well. We present the stacked spectra from all subsamples in Fig-

Table 2. Subsamples Properties

ID	N	$\langle z \rangle$	z_{lower}	z_{upper}
(1)	(2)	(3)	(4)	(5)
#1	7	7.12	7.045	7.179
#2	6	7.44	7.197	7.781
#3	7	8.28	7.834	8.683
#4	7	9.91	8.763	11.40

NOTE— (1): ID for a subsample. (2): Size of a subsample. (3): Representative (i.e., the average) redshift of galaxies in a subsample. (4): Lowest redshift of galaxies in a subsample. (5): Highest redshift of galaxies in a subsample.

ure 5. Fluxes of the composite spectra are normalized at the rest frame 1450 Å. As shown in Figure 5, we can clearly see the softening break feature at the rest-frame 1216 Å towards the high redshift, suggesting increasing Ly α damping wing absorptions due to increasing $x_{\text{H I}}$ with redshifts.

3.3. Spectral Fitting

To further confirm the redshift evolution of Ly α damping wing absorption, we measure the IGM Ly α damping wing absorption. At the same time, we also consider the effect on the shape of Lyman break from the intrinsic galaxy's property such as stellar mass. Because the interpretation of spectral fitting using stacked spectrum would be complicated as spectra of galaxies with different physical properties are mixed together, we fit individual galaxy's spectrum individually and consider marginalized posterior distributions for damping wing parameters. For the fitting, we use the observed spectra in the wavelength range above the rest-frame 1000 Å without masking Ly α emission. We included data at shorter wavelength value than the rest-frame Ly α (i.e., 1216 Å) to consider the effect of higher flux transmission due to large surrounding ionized bubble. We generate model spectrum in forward modeling manner to compare the observed spectrum to model spectrum. We first generate the model spectra incorporating stellar and nebular emission. We also add Ly α emission prior to applying the Ly α absorption. Then, we apply Ly α absorption to the intrinsic model spectrum. After applying Ly α absorptions, we smooth the model spectra

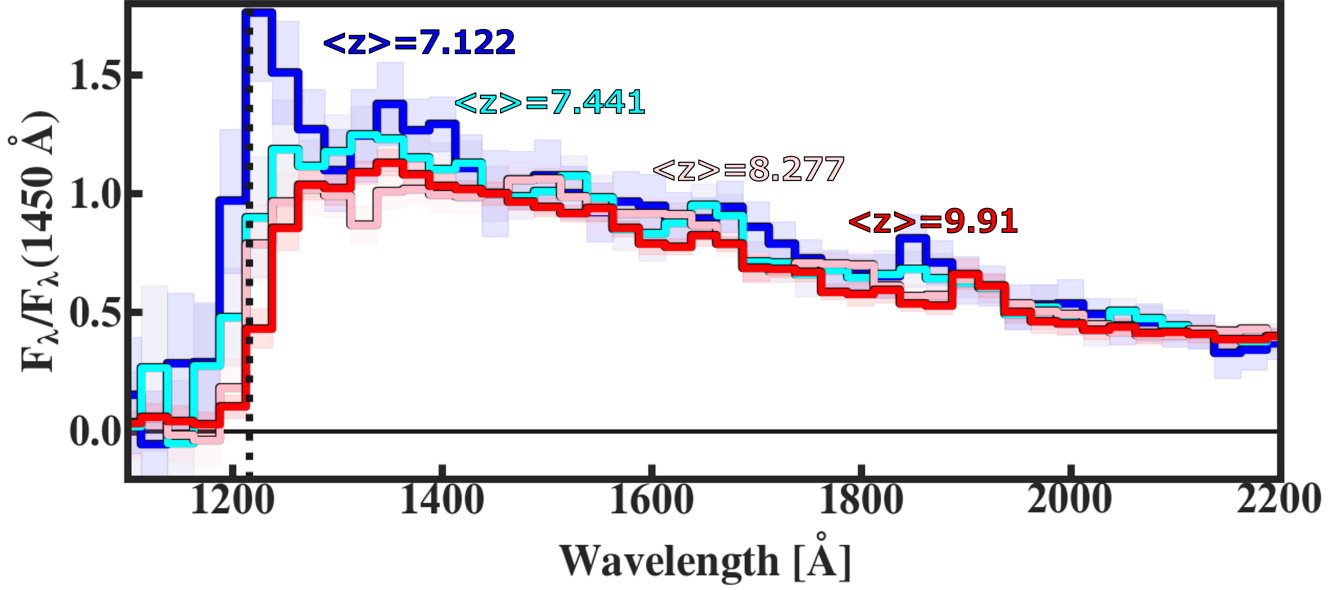


Figure 5. Stacked spectra of our galaxy sample binned into four redshift bins. The dark blue, pale blue, pink, and red solid lines (shades) represent the stacked spectra (uncertainties) for the redshift bins at $\langle z \rangle = 7.12, 7.44, 8.28$, and 9.91 , respectively. The wavelength is shifted to the rest frame and the flux is normalized at $1450 \text{ \AA}$ at which the IGM absorption have negligible impact on the spectrum. The black vertical dotted line represent the position of the rest-frame $1216 \text{ \AA}$. The horizontal solid line indicates the position of the flux value at zero.

Table 3. Prior Distributions for Fitting Parameters

$\log M_*$	$\log(SFR_{i+1}/SFR_i)$	$\tau_{5500\text{\AA}}$	$\log U$	x_{HI}	$\log R_b$	$\log N_{\text{HI}}$	$\Delta_{v,\text{DLA}}$	σ_v	$\log EW_{\alpha,0}$
($M_\odot$)	($i = 1, 2, 3, 4, 5$)				(cMpc)	(cm^{-2})	(km s^{-1})	(km s^{-1})	($\text{\AA}$)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
U(6,11)	t(0,0.3,2)	U(0,2)	U(-6,2)	U(0,1)	U(-1,3)	U(10,23)	N(0,300)	U(0,100)	U(-3,3)

NOTE— (1): Total stellar mass. (2): Star formation rate ratio for six star formation history bin. (3): Optical depth at $5500 \text{ \AA}$ for dust attenuation law by Calzetti et al. (2000). (4): Ionization parameter. (5): Neutral hydrogen fraction. (6): Ionized bubble radius. (7): Neutral hydrogen column density for CGM. (8): Velocity offset for CGM relative to galaxy systemic redshift. (9): Velocity dispersion for CGM. (10): Intrinsic $\text{Ly}\alpha$ equivalent width. $N(\mu, \sigma)$ is a normal distribution with mean μ and variance σ^2 . $U(a, b)$ is a uniform distribution between a and b . $t(\mu, \sigma, \nu)$ is a student's t distribution with mean μ , scale factor σ , and degree of freedom ν .

with line spread function of PRISM provided by STScI to account for the instrumental broadening.²

We use a SED fitting code based on **prospector** (Johnson et al. 2021) to generate model galaxy spectra with various stellar population and nebular properties. We use the Binary Population and Spectral Synthesis (BPASS; Eldridge et al. 2017) model as an isochrone library. We allow different non-parametric star formation

history (SFH), stellar mass (M_*), ionization parameter ($\log U$), and dust attenuation. We adopt the star formation history in the same manner as Harikane et al. (2023). We also allow different stellar mass in the range of $6 < M_* < 11$, which fully covers the mass range of $z > 7$ galaxies estimated in Nakajima et al. (2023) and Harikane et al. (2023). For the dust attenuation, we apply the attenuation law by Calzetti et al. (2000) and use an optical depth at $5500 \text{ \AA}$ as a free parameter. We also include nebular emission continuum and lines, but we treat $\text{Ly}\alpha$ emission line separately as we describe later in this section. We fix stellar/nebular metallicity to 10%

² <https://jwst-docs.stsci.edu/jwst-near-infrared-spectrograph/nirspec-instrumentation/nirspec-dispersers-and-filters>

solar metallicity as suggested for JWST high- z galaxy by recent work of Nakajima et al. (2023). We set the same prior distributions for free stellar/nebular parameters as adopted in Nakajima et al. (2023).

For Ly α absorption, we consider Ly α absorption by both IGM and circum-galactic medium (CGM). We define the intrinsic Ly α emission as the Ly α emission before encountering CGM/IGM absorption. For intrinsic Ly α emission, we let the equivalent width of the intrinsic Ly α emission line ($\log EW_{\alpha,0}$) free in the range of $-3 < \log EW_{\alpha,0}/\text{\AA} < 3$. In this way, we can consider both extreme and negligible (i.e., low escape fraction) contribution of Ly α emission to the shape of Lyman break. We discuss about the validity of Ly α emission line strengths in Section 4.2. We fix the velocity offset and dispersion of the Ly α emission line to +200 km/s and 250 km/s, respectively. We also discuss about how this assumption on the velocity offset affect the result in Section 4.5. For Ly α absorptions, we adopt the formulation given by equation 2 for the IGM absorption with $x_{\text{H I}}$ and $\log R_b$ as free parameter. For the H I absorption by CGM, we consider the Voigt profile with different neutral hydrogen column density ($\log N_{\text{H I}}$), velocity dispersion (σ_v), and velocity offset relative to the systemic redshift ($\Delta_{v,\text{DLA}}$). We summarize our fitting parameters and corresponding prior distributions in Table 3.

We run Markov Chain Monte Carlo (MCMC) method to obtain the posterior probability distribution functions (PDFs) of free parameters. To take an account of variance in galaxy's intrinsic spectral properties in estimating Ly α absorption parameters, we simultaneously fit for the stellar, nebular, and absorptions. We use a package created based on `emcee` (Foreman-Mackey et al. 2013) called `ptemcee` (Vousden et al. 2021) for MCMC. `ptemcee` incorporates parallel tempering technique that realizes faster sampling convergence in case of multimodal posterior probability distribution functions. We run four temperature parallel tempering MCMC sampler of 80 walkers for each temperature with 5000 steps. We discard first 1000 steps of each walker to diminish the dependence of posterior PDFs to initial parameter distributions.

We infer the best-fit parameters from the posterior PDFs. Because a median value does not well represent the skewed posterior distribution, we define the best-fit parameter by the mode (i.e., a peak of the posterior distribution) and uncertainty by 68th percentile highest posterior density interval (HPDI; i.e., a narrowest interval containing 68%) of the posterior PDFs. To obtain mode of 1D marginalized posterior PDF for each parameter, we smooth the distribution by Kernel Density Estimation smoothing using `scipy` package (Virtanen

et al. 2020) and choose the mode within an HPDI of 1D posterior distribution. We also calculate the posterior probability distribution within each redshift bin (hereafter, binned posterior PDF) in the following manner:

$$\begin{aligned} p(\theta|\mathbf{D}) &= \frac{p(\mathbf{D}|\theta)p(\theta)}{p(\mathbf{D})} \\ &= \left[\prod_i^N \frac{p(D_i|\theta)p(\theta)}{p(D_i)} \right] p^{1-N}(\theta) \\ &= \left[\prod_i^N p(\theta|D_i) \right] p^{1-N}(\theta). \end{aligned} \quad (4)$$

Here, $p(\cdot)$, θ , N , and D_i ($\mathbf{D} \equiv \{D_i\}_{i=1,\dots,N}$) represent the probability density, free parameters, number of galaxy in each redshift bin, and i -th (all set of) galaxy spectral data in each bin, respectively. In equation 4, we assume

$$\begin{aligned} p(\mathbf{D}) &= \prod_i^N p(D_i), \\ p(\mathbf{D}|\theta) &= \prod_i^N p(D_i|\theta). \end{aligned} \quad (5)$$

We utilize the posterior PDF for individual galaxies (i.e., $p(\theta|D_i)$) obtained with MCMC to calculate the binned posterior PDFs. We obtain the best-fit value for each redshift bin in the similar manner as for the individual galaxies.

3.4. Fitting Results

In Figure 6, we show the binned posterior PDF of the parameter of $x_{\text{H I}}$ and $\log R_b$ for each redshift bin. The solid (dashed) contour represents the 68-(90)-th percentile range for 2D marginalized PDFs. The top two panels in corner diagrams represent the marginalized probability density function of the $x_{\text{H I}}$ and $\log R_b$. We summarize the best-fit parameters in Table 4. For the results of the #4 bin, we obtain very high $x_{\text{H I}}$ and small R_b values (i.e., $x_{\text{H I}} \sim 1$, $R_b < 1$ cMpc) as similar as the ones reported for a $z = 10.17$ galaxy in Hsiao et al. (2023). For the $z \sim 7$ galaxies, our $x_{\text{H I}}$ is not well constrained as one can see from the marginalized PDF. The reason for tighter constraint on $\log R_b$ compared to that for $x_{\text{H I}}$ could be attributed to the strong dependence of $\log R_b$ value to the position of observed Lyman break relative to the systemic redshift. Moreover, the reasons for having sharp peaks for $\log R_b$ compared with that for $x_{\text{H I}}$ could be attributed to that Ly α damping wing shapes are more sensitive $\log R_b$ than for $x_{\text{H I}}$ as $\log R_b$ have uniform prior distributions based on logarithmic form, unlike the linearly uniform prior distributions for $x_{\text{H I}}$. Logarithmically changing $\log R_b$ values have a little impact on the shape of model spectra, in another

Table 4. Best-fit Estimate for x_{HI} and $\log R_b$

ID	$\langle z \rangle$	x_{HI}	$\log R_b$ [cMpc]
(1)	(2)	(3)	(4)
#1	$7.12^{+0.06}_{-0.08}$	$0.53^{+0.18}_{-0.47}$	$1.67^{+0.14}_{-0.16}$
#2	$7.44^{+0.34}_{-0.24}$	$0.65^{+0.27}_{-0.34}$	$1.53^{+0.34}_{-1.48}$
#3	$8.28^{+0.41}_{-0.44}$	$0.91^{+0.09}_{-0.22}$	$0.70^{+0.47}_{-1.04}$
#4	$9.91^{+1.49}_{-1.15}$	$0.92^{+0.08}_{-0.10}$	$-0.69^{+0.89}_{-0.24}$

NOTE— (1): ID for each subsample. (2): Mean redshift and the lower/upper boundary of a subsample. (3): Mode value and upper/lower limit of HPDI of x_{HI} for the stacked marginalized PDFs in each redshift bin. each redshift bin. (4): Mode value and upper/lower limit of HPDI of $\log R_b$ for the stacked marginalized PDFs in each redshift bin.

word the likelihood values do not change dramatically, when the $\log R_b$ is considerably small. This is reflected on the low and flat posterior probability values at small $\log R_b$ range and sharp peaks around the best-fit values in Figure 6. We also confirm similar behaviors for posterior probability distribution for $\log R_b$ for individual galaxies. We note that narrowing down the range of flat priors for $\log R_b$ does not affect the best-fit value as it is defined by the mode, but the lower bound for uncertainty would increase if we adopt higher value for the lower bound of the flat prior.

In Figure 7, we also present examples of our spectrum fitting for individual objects. From the top to bottom panels, we show the fitting results for CEERS_1019, CEERS2_588, and 00021842, respectively. For CEERS_1019, strong IGM absorption with $x_{\text{HI}} = 0.91^{+0.09}_{-0.45}$ and an ionized bubble size of $\log R_b/\text{cMpc} = 0.82^{+0.61}_{-0.45}$ explains moderately softened break feature around the rest-frame 1216 Å of the observed spectrum. For CEERS2_588, in addition to strong IGM absorption (i.e., $x_{\text{HI}} \sim 1$) and small ionized region (i.e., $\log R_b/\text{cMpc} < 0$), Ly α damping wing absorption by CGM with H I column density of $\log N_{\text{HI}}/\text{cm}^{-2} > 22$ surrounding the galaxy explains strongly softened break feature around the rest-frame 1216 Å. In contrast, the sharp and blue-shifted break feature at the rest-frame 1216 Å for 00021842 can be explained with Ly α emission together with high Ly α transmission environment realized with small IGM absorption (i.e., $x_{\text{HI}} \sim 0$) and a large ionized bubble (i.e., $\log R_b/\text{cMpc} > 1$).

4. DISCUSSION

4.1. Ly α Absorption by CGM

In Figure 8, we summarize the best-fit estimation of neutral hydrogen column density of H I gas in CGM by redshift. In the same figure, we also present the measured neutral hydrogen column density of $z \sim 0$ metal-poor galaxies (Hernandez et al. 2020), $z \sim 0$ green pea galaxies (McKinney et al. 2019), gamma ray bursts (Tanvir et al. 2019), and recent $z > 8$ bright galaxies (Heintz et al. 2023). Measured neutral hydrogen column density spans up to the range of damped Ly α absorbers (DLA; $N_{\text{HI}} > 2 \times 10^{20}$). Moreover, neutral hydrogen column density reaches up to $\gtrsim 10^{22} \text{ cm}^{-2}$ for some objects, suggesting a existence of extreme reservoir of neutral hydrogen gas around some of the $z > 7$ galaxies as suggested by Heintz et al. (2023). We compare the neutral hydrogen column density measurement of overlapping three galaxy between our sample and Heintz et al. (2023)’s and find that our measurements are consistent within errorbars with Heintz et al. (2023)’s. This consistency further solidifies the validity of our analysis and the existence of high- z extreme DLA.

4.2. Escape Fraction

As the existence of extremely high neutral hydrogen column density CGM is suggested around some of the $z > 7$ bright galaxies, we consider how much Ly α emission can escape to the IGM (i.e., Ly α escape fraction $f_{\text{esc,Ly}\alpha}$). We estimate $f_{\text{esc,Ly}\alpha}$ by comparing the Ly α emission flux after the absorption by CGM and the intrinsic Ly α emission flux. For the intrinsic Ly α emission flux, we use cloudy prediction calculated by Byler et al. (2017) as Prospector adopts. We note that we do not apply IGM absorption to calculate $f_{\text{esc,Ly}\alpha}$. We also estimate $E(B - V)$ using sampled posterior PDFs of optical depth at 5500 Å by dust attenuation. In Figure 9, we plot estimated $E(B - V)$ and $f_{\text{esc,Ly}\alpha}$ and compare with local relation discussed in Atek et al. (2014). As shown in Figure 9, best-fit value for our $f_{\text{esc,Ly}\alpha}$ are below unity, which means that Ly α emission strength applied in our analysis does not violate prediction from photoionization models. Moreover, our estimates for $f_{\text{esc,Ly}\alpha}$ are roughly consistent with the local relation between $E(B - V)$ and $f_{\text{esc,Ly}\alpha}$, although some of our $f_{\text{esc,Ly}\alpha}$ values are smaller than the local trend. Because emission line flux is smoothed in low-resolution PRISM spectra, $f_{\text{esc,Ly}\alpha}$ cannot be well constrained for smaller values. Moreover, galaxy sample used to derive the relation in Atek et al. (2014) is partially selected based on the Ly α emission detection. In contrast, we construct our galaxy sample based on the brightness of continuum (i.e., UV magnitude), so it is natural to have our $f_{\text{esc,Ly}\alpha}$

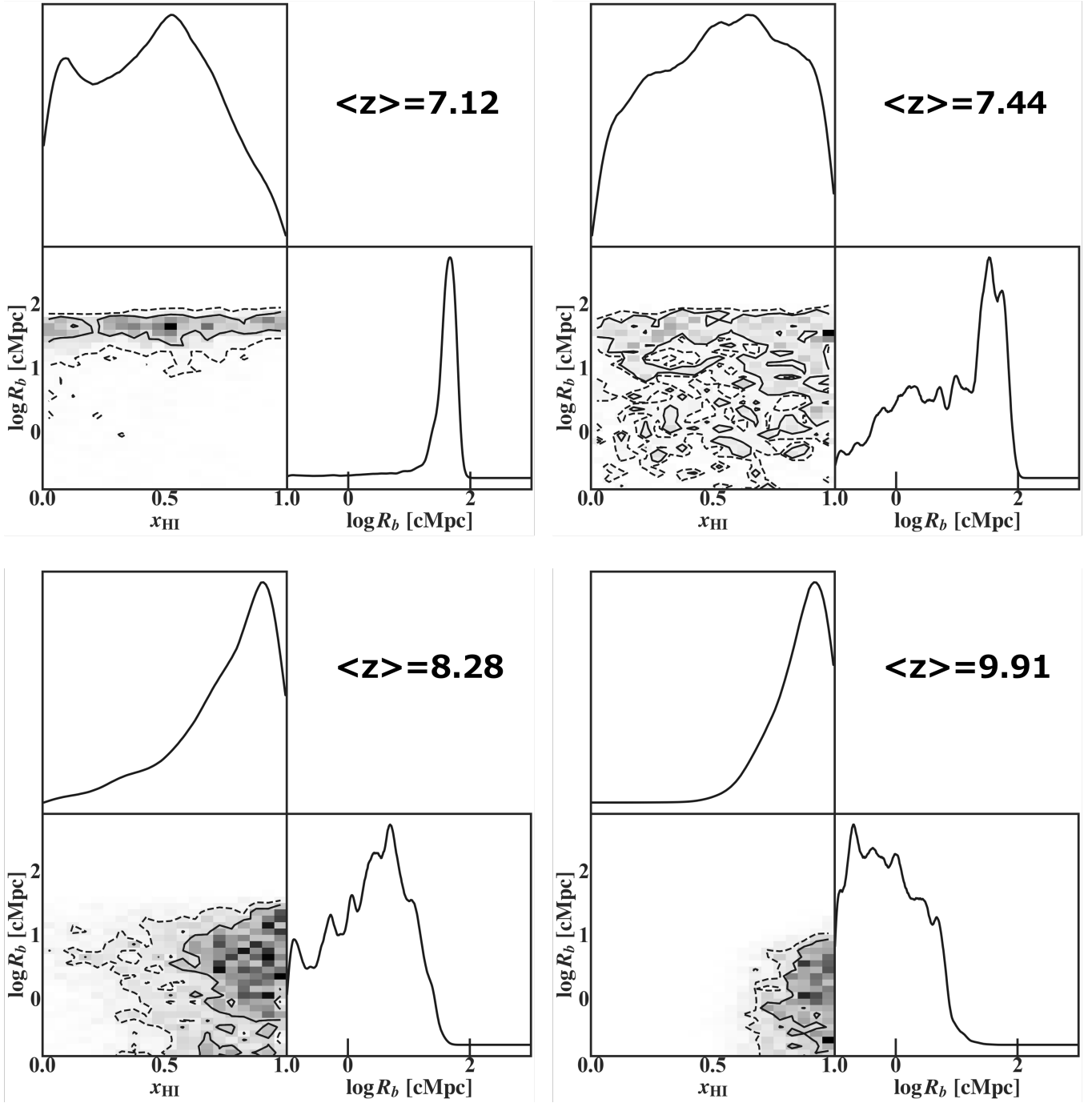


Figure 6. The posterior probability distribution function (PDF) of x_{HI} and $\log R_b$ for each subsample. The results for #1 to #4 bin are shown in order of top left to right bottom. The darkness in the 2D-marginalized PDFs represent the probability density. The solid and dotted contours represent the 68 and 90-th percentiles, respectively. Top panel corresponds to the 1D marginalized PDF for each parameter. The presented PDFs are smoothed by gaussian kernel.

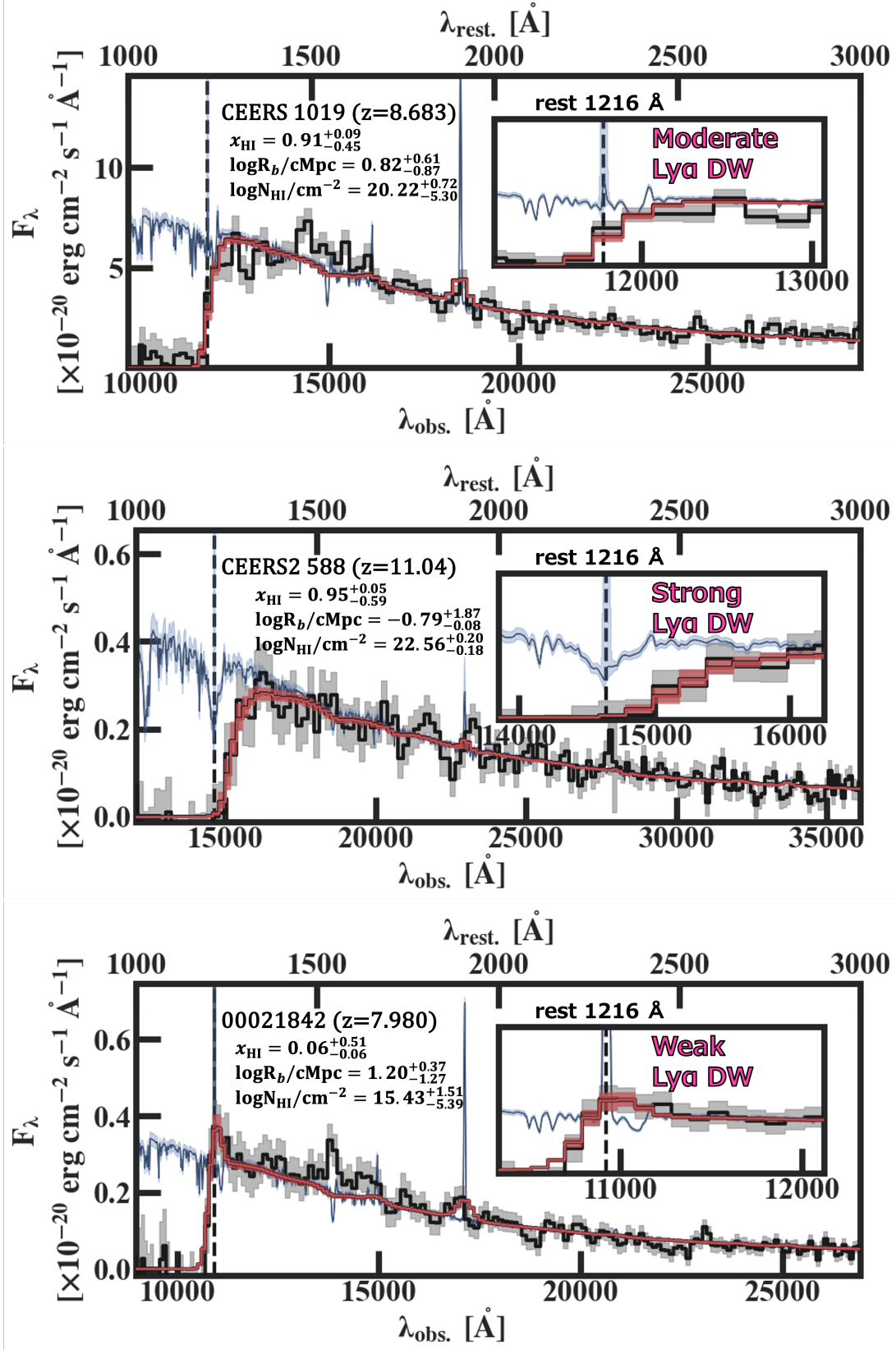


Figure 7. Fitted spectrum for the case of moderate, strong and weak Ly α damping wing absorption feature seen in CEERS_1019, CEERS2_588, and 000012842 from the top to bottom panels, respectively. The black line and shade represent the observed spectrum and its error. The red (blue) line and shade represent the model spectrum and its uncertainty after (before) Ly α absorptions applied and smoothed with PRISM resolution. The black dotted line represent the rest-frame $1216 \text{ \AA}$ at the systemic redshift. The inset panel shows zoom-in view of the spectrum around the rest-frame $1216 \text{ \AA}$. The best-fit values and uncertainties for x_{HI} , $\log R_b$, and $\log N_{\text{HI}}$ for each objects are also written inside each panel. The x-axes at the top and bottom of each panel represent the observed and rest-frame wavelength, respectively.

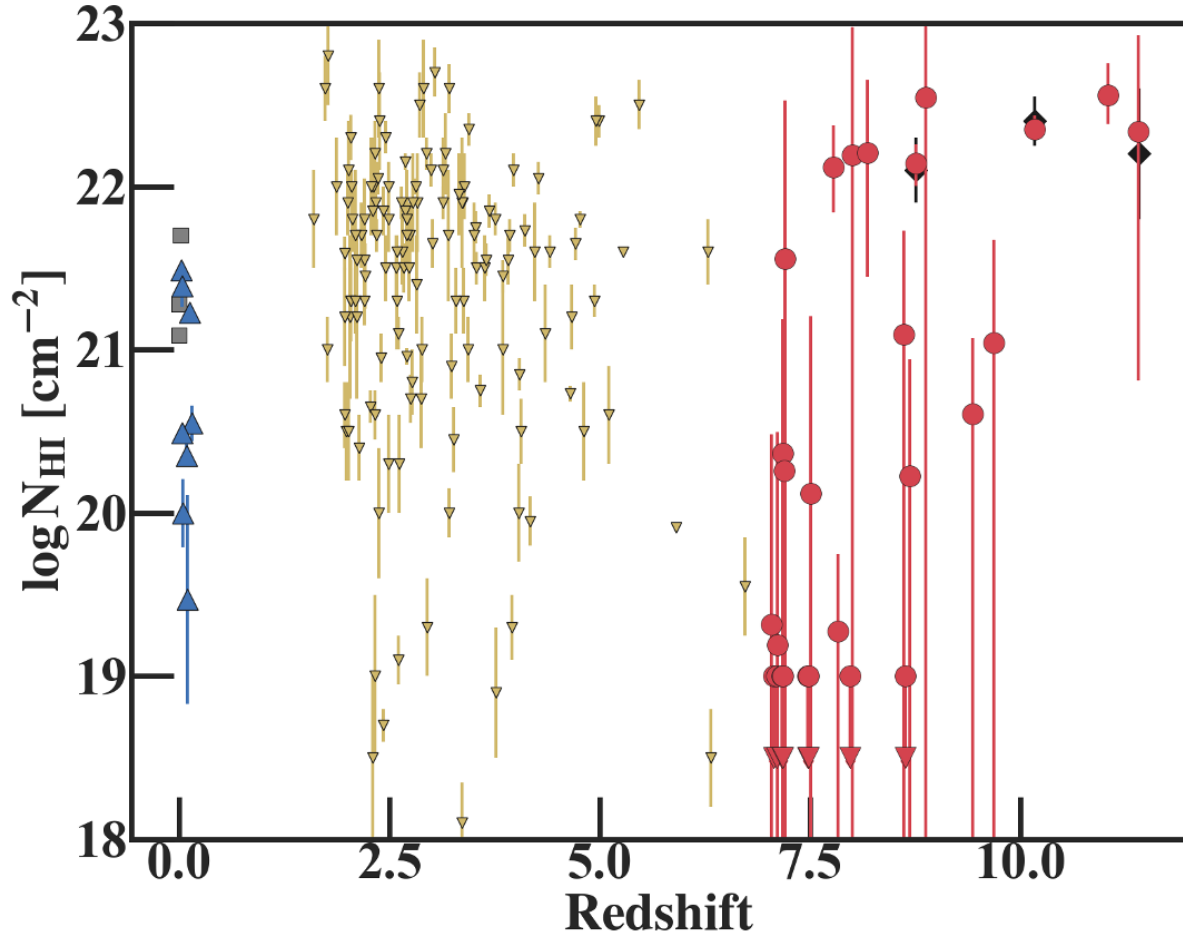


Figure 8. Neutral hydrogen column density of Ly α absorption by H I gas at different redshifts. The red circles (bars) represent $\log N_{\text{HI}}$ measurements (uncertainties). The blue upward triangles, gray squares, and yellow downward triangles, and black diamonds represent the $\log N_{\text{HI}}$ measurements (uncertainties) of $z \sim 0$ green pea galaxies (McKinney et al. 2019), $z \sim 0$ metal poor galaxies (Hernandez et al. 2020), gamma ray bursts (Tanvir et al. 2019), and $z > 9$ Lyman break galaxies (Heintz et al. 2023), respectively.

to lie on or below the local trend of Atek et al. (2014). In any case, we have confirmed that our treatment of Ly α emission is physical and consistent with observational trend.

4.3. Cosmic Reionization History

In Figure 10, we plot our x_{HI} constraints and the uncertainties in red circle and bars, respectively. We also show literature values of x_{HI} at different redshift including the constraint suggested by the electron scattering from the cosmic microwave background observations (Planck Collaboration et al. 2020b) and Ly α damping wing measurements of QSO spectra (Schroeder et al. 2013; Davies et al. 2018; Greig et al. 2019; Wang

et al. 2020), GRB spectra (Totani et al. 2006, 2014), and recently obtained JWST galaxy spectra in others works (Curtis-Lake et al. 2023; Hsiao et al. 2023). Optical depth measurements using QSOs Gunn Peterson troughs around $z \sim 5 - 6$ place the constraints on the end of the epoch of reionization (Fan et al. 2006). We additionally show other x_{HI} constraints from galaxy observations including the ones inferred from Ly α luminosity function (Ouchi et al. 2010; Konno et al. 2014; Inoue et al. 2018; Konno et al. 2018; Morales et al. 2021; Goto et al. 2021), Ly α emitter (LAE) clustering analysis (Ouchi et al. 2018), Ly α +Ly β dark fraction/gaps (Mesinger et al. 2015; Jin et al. 2023; Zhu et al. 2022), and Ly α emission lines of Lyman-break galaxies

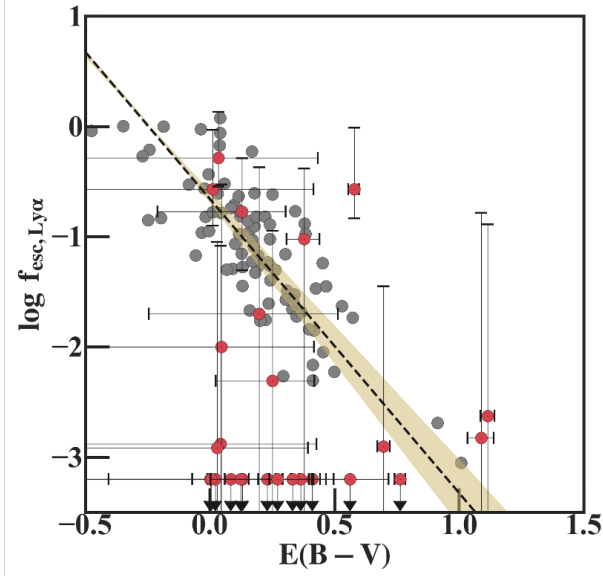


Figure 9. Relation between $E(B - V)$ and $\text{Ly}\alpha$ escape fraction. The red circle and bars (arrows) represent best-fit and errors (upper limits) for our galaxy sample. The grey circle represent the measurement of $z \sim 0$ galaxies presented in left panel of Figure 10 from Atek et al. (2014). The black dotted line and yellow shades represent the best-fit relation and its errors between $E(B - V)$ and $f_{\text{esc}, \text{Ly}\alpha}$ for $z \sim 0$ galaxies as presented in Atek et al. (2014).

(Mesinger et al. 2015; Hoag et al. 2019; Mason et al. 2019; Jung et al. 2020; Whitler et al. 2020; Bruton et al. 2023; Morishita et al. 2023). For the x_{HI} estimate from Curtis-Lake et al. (2023), we adopt the values shown in Figure 3 of the corresponding paper.

We overplot three distinct evolutionary models by Naidu et al. (2020), Finkelstein et al. (2019), and Ishigaki et al. (2018) (hereafter, Naidu model, Finkelstein model, Ishigaki model, respectively) in orange dashed, blue solid, and purple dotted lines, respectively. We adopt Model II described in Naidu et al. (2020) as Naidu model. Note that the difference between redshift evolution of x_{HI} in Model I and II in Naidu et al. (2020) is insignificant compared to relatively large errorbars for current x_{HI} measurements. These evolutionary models differ in the prescription for the escape fraction f_{esc} and the faint end slope α of the UV luminosity functions. Naidu model assumes that f_{esc} correlates with the star formation surface density Σ_{SFR} with a shallow faint end slope (i.e., $\alpha > -2$), resulting in a late and rapid cosmic reionization history driven by a small number of bright galaxies. Finkelstein model assumes that f_{esc} anti-correlates with halo mass M_{halo} and a steep

faint end slope (i.e., $\alpha < -2$), resulting in an extended x_{HI} evolution driven by many faint galaxies. Ishigaki model assumes the constant f_{esc} value at $\simeq 0.17$ with a steep UV faint end slope (i.e., $\alpha < -2$). Ishigaki model results in a moderately late cosmic reionization consistent with the one suggested by an electron scattering of the cosmic microwave background. We summarize the prescriptions for each model in Table 5. As shown in Figure 10, the best estimate of our x_{HI} consistent with all three evolutionary models and the constraints inferred from the electron scattering of CMB.

In Figure 10, we also overplot the x_{HI} values at $z \approx 7$ estimated by Umeda (2023) from $\text{Ly}\alpha$ luminosity function and clustering properties of LAEs as pink circle and square, respectively. Umeda (2023) derive $\text{Ly}\alpha$ luminosity function and LAE spatial angular correlation function using the $z = 6 - 7$ photometric LAE catalogs constructed by Ono et al. (2021) from the Public Data Release 2 data of the Hyper Suprime-Cam Subaru Strategic Programs (Aihara et al. 2019). Umeda (2023) estimate x_{HI} from $\text{Ly}\alpha$ luminosity function and angular correlation function of LAEs in a similar manner as Konno et al. (2018), Ouchi et al. (2018), and Goto et al. (2021). The $z = 6 - 7$ LAE catalog of Ono et al. (2021) is the largest in sample size with a total of 3561 galaxies, thus the x_{HI} estimates from Umeda (2023) have advantages in terms of the reduced statistical uncertainty for estimated $\text{Ly}\alpha$ luminosity function and angular correlation functions. For the x_{HI} estimate from $\text{Ly}\alpha$ luminosity function, Umeda (2023) first compared the luminosity density at $z = 5.7$ and $z = 6.6$ to derive IGM transmission of $\text{Ly}\alpha$. Then, Umeda (2023) estimate x_{HI} from the IGM transmission using the analytical relation between IGM transmission of $\text{Ly}\alpha$, x_{HI} , and the ionized bubble size from Dijkstra (2014) and the semi-numerical relation between x_{HI} and R_b from Furlanetto & Oh (2005). For the x_{HI} estimate from LAE clustering analysis, Umeda (2023) compared the derived galaxy bias evolution from $z = 5.7$ to $z = 6.6$ to the semi-numerical prediction of galaxy bias at $z = 6.6$ with different x_{HI} by Furlanetto et al. (2006). By comparing the x_{HI} estimate at $z > 7$ with the one at $z < 7$ estimated by Umeda (2023), we can confirm smooth monotonic increase of x_{HI} towards the higher redshift. Other x_{HI} constraints at $z > 7$ from the literature are mostly consistent with our measurements within error. To summarize, our x_{HI} measurements at $z > 7$ agree with various constraints, including the ones previously inferred from the luminosity function and clustering properties of LAEs at $z < 7$, the electron scattering of CMB, and the evolution of UV luminosity function assuming a constant $f_{\text{esc}} \sim 0.2$.

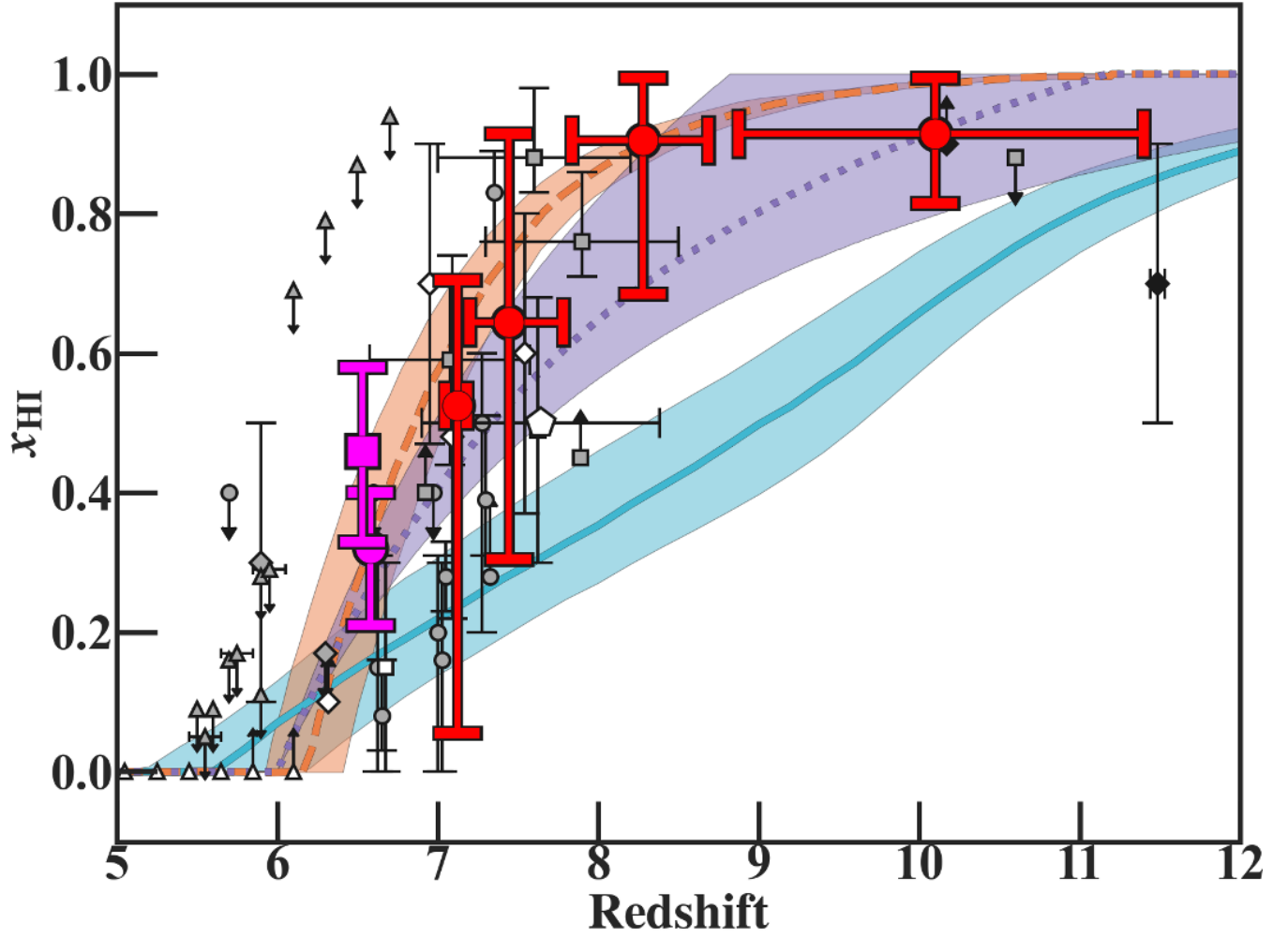


Figure 10. Redshift evolution of x_{HI} . The red circle and bars represent the x_{HI} estimates and the uncertainties from our work. The pink circle (square) and bars show the x_{HI} estimate and uncertainty from $z = 6 - 7$ Ly α luminosity function (LAE clustering analysis) with data obtained with Subaru/Hyper Suprime Cam (Umeda 2023). The orange dashed, blue solid, and purple dotted line and the shaded with corresponding colors are the model and its uncertainty for the redshift evolution of x_{HI} predicted by Naidu et al. (2020), Finkelstein et al. (2019), and Ishigaki et al. (2018), respectively. The grey circles and white squares represent the x_{HI} estimation from Ly α luminosity function (Ouchi et al. 2010; Konno et al. 2014; Inoue et al. 2018; Konno et al. 2018; Morales et al. 2021; Goto et al. 2021) and LAE clustering analysis (Ouchi et al. 2018), respectively. The white and grey triangles represent the x_{HI} estimation by QSOs Gunn Peterson troughs (Fan et al. 2006) and Ly α +Ly β dark fraction/gaps (Mesinger et al. 2015; Jin et al. 2023; Zhu et al. 2022), respectively. The grey and white diamonds represent the x_{HI} estimation from Ly α damping wing absorption of QSOs (Schroeder et al. 2013; Davies et al. 2018; Greig et al. 2019; Wang et al. 2020) and GRBs (Totani et al. 2006, 2014). The black diamonds represent the x_{HI} estimation from Ly α damping wing absorption profile of individual UV spectrum of galaxies (Curtis-Lake et al. 2023; Hsiao et al. 2023). The grey squares represent the x_{HI} estimation from Ly α of Lyman-break galaxies (Mesinger et al. 2015; Hoag et al. 2019; Mason et al. 2019; Jung et al. 2020; Whitler et al. 2020; Bruton et al. 2023; Morishita et al. 2023). The white pentagon represents the x_{HI} estimation from the cosmic microwave background observations (Planck Collaboration et al. 2020b).

Table 5. $x_{\text{H I}}$ Evolution Models

Model	Naidu	Ishigaki	Finkelstein
f_{esc}	$\propto \Sigma_{\text{SFR}}^{0.4}$	0.17	$\propto M_{\text{halo}}^{-1}$
α	> -2	< -2	< -2

NOTE—Naidu, Ishigaki, and Finkelstein models corresponds to $x_{\text{H I}}$ evolution model presented in Naidu et al. (2020), Ishigaki et al. (2018), and Finkelstein et al. (2019), respectively. We adopt Model II of the Naidu et al. (2020) as Naidu model.

4.4. Bubble Size Evolution

Beside $x_{\text{H I}}$, the size of the ionized bubble is another tracer of cosmic reionization scenarios. Various studies suggest that in the case of inhomogeneous cosmic reionization driven by galaxies, the ionized region forms around the galaxies and their typical size correlates with $x_{\text{H I}}$ (e.g., Ciardi & Madau 2003; Furlanetto & Oh 2005; Mesinger & Furlanetto 2008a,b; Lu et al. 2023). The size of ionized bubble at the EoR has been inferred from the Ly α emission line profiles and equivalent widths (e.g., Mason & Gronke 2020; Endsley & Stark 2022; Hayes & Scarlata 2023; Jung et al. 2023; Saxena et al. 2023). We instead use Ly α damping wing profiles to investigate R_b and its evolution with the redshift. We use the best-fit parameters presented in Section 3 to see the relation between R_b and $x_{\text{H I}}$. Figure 11 shows the estimated R_b values for each redshift bin from our analysis in Section 3 together with the relation between $x_{\text{H I}}$ and R_b predicted from analytical model by Furlanetto & Oh (2005) in black dotted lines. Together with our measurements, we show $x_{\text{H I}}$ and R_b constrained via damping wing absorption analysis in Hsiao et al. (2023). We confirm that our constraint for the redshift bin of $\langle z \rangle = 9.91$ is consistent with $x_{\text{H I}}$ and R_b measurements for $z = 10.17$ object by Hsiao et al. (2023). In contrast, the recent bubble size measurement by Fujimoto et al. (2023b) for $z = 8.51$ object suggests a bubble size of $R_b = 71.9 \pm 1.0$ cMpc. This bubble size value is about a dex larger than our constraint at the redshift bin of $\langle z \rangle = 8.28$. As Fujimoto et al. (2023b) discusses in their paper, this large offset of bubble size measurement could be physically attributed to specifically strong ionizing radiation by two possible AGN candidates with only a separation of 380 kpc within the same ionized bubble. We also plot the $x_{\text{H I}}$ and R_b relation for the rapid and gradual cosmic reionization history predicted from a semi-numerical simulations by Lu et al. (2023) in solid and dashed lines, respectively. Lu et al. (2023)’s predictions take into account of the overlap of the ionized regions formed around

UV bright ($M_{\text{UV}} < -20$) galaxies that are comparable to those for the majority of galaxies in our sample. Incorporation of the overlaps between ionized yields systematically larger R_b at each $x_{\text{H I}}$ compared with the ones predicted by Furlanetto & Oh (2005), which do not consider the overlaps. As shown in Figure 11, our R_b estimate at each $x_{\text{H I}}$ is up to few dex larger than the one predicted by theoretical relations of Furlanetto & Oh (2005), shown as the dotted line. However, our R_b estimates are consistent with in errorbars with the ones predicted by Lu et al. (2023) that consider the overlaps of ionized regions, confirming the importance of considering the overlaps to characterize the typical ionized bubble shapes during the EoR. Our inferred ionized bubble size is unexpected from the previous Ly α forest measurement. Ly α optical depth calculated based on our inferred $z \sim 7 - 7.5$ ionized bubble radius is only found at $z \lesssim 3$ by QSO Ly α forest measurements by Becker et al. (2013). However, the Ly α optical depth measurement of Becker et al. (2013) does not incorporate contributions from high Ly α transmission at the proximity zone of the background QSO. Because our galaxies could be located at the overdensed region with a number of ionizing sources, we must consider the proximity effect to have a fair comparison between our $z > 7$ Ly α transmission estimate and those at $z < 7$. When we compare our bubble size estimates to the previous measurement from QSO observations, our best-fit bubble radii estimates (i.e., $\sim 10^2$ cMpc) at $z < 8$ are even larger than the size of proximity zone around the brightest $z \sim 6$ QSOs (i.e., $\lesssim 10$ cMpc) (e.g., Eilers et al. 2018; Keating et al. 2023). Despite our bubble size estimates and the QSO proximity zone size does not have strong discrepancy because of the large errorbars for our measurement, one possibility to explain QSOs having smaller ionized region surrounding it than galaxy is that the QSO have shorter duty cycle than that of galaxies. Because gas needs to be irradiated by ionizing radiation long enough to form proximity zone, short QSO lifetime ($< 10^5$ yr) has been proposed to explain small proximity observed at $z \sim 6$ QSOs (Davies et al. 2018). The steady supply of ionizing photons through continuous star formation in galaxies could possibly enlarge the ionized bubble size more efficiently than young or short lived QSOs discovered at $z > 6$. Precise R_b measurements in the future would be useful to distinguish different cosmic reionization scenarios (specifically on the formation process of ionized regions) that yield different $x_{\text{H I}}$ and R_b relations.

4.5. Possible Systematics

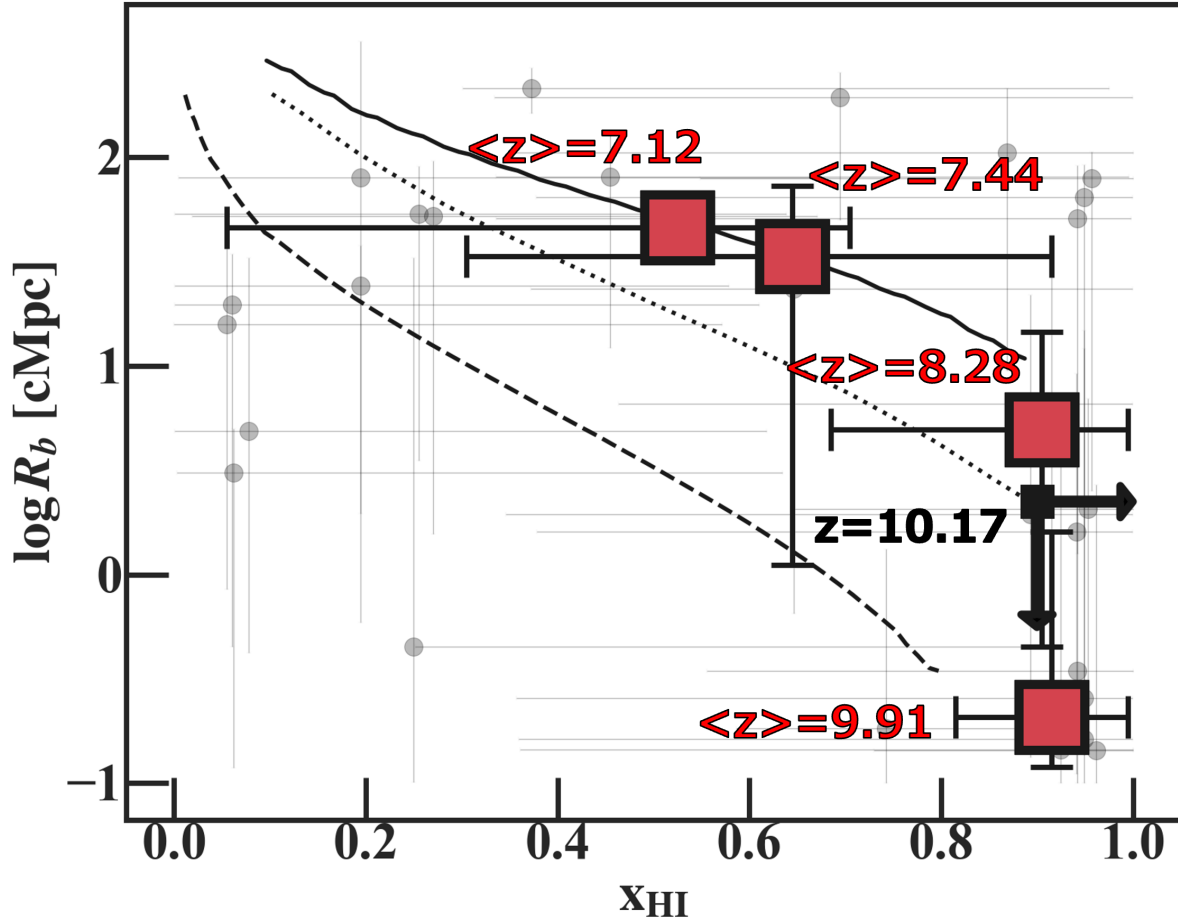


Figure 11. x_{HI} and R_b estimates from different redshift bins. The red squares and bars represent our estimates and the errors, respectively, of $(x_{\text{HI}}, \log R_b)$ at the redshift bin #1, #2, #3, and #4. Black square and arrows corresponds to 1σ upper and lower limits of x_{HI} and R_b , respectively, on MACS0647-JD ($z = 10.17$) by Hsiao et al. (2023). Faint grey circles and bars represent the best-fit value and its error for individual galaxies. Solid and dashed lines correspond to the theoretical prediction by Lu et al. (2023) on the characteristic ionized bubble radii at different x_{HI} assuming rapid and gradual cosmic reionization considering overlapping ionized regions, respectively. The dotted line corresponds to the analytical relation between characteristic ionized bubble radii and x_{HI} without considering overlapping ionized regions (Furlanetto & Oh 2005).

Although we consider the ionized bubbles surrounding the galaxies and H I absorption by CGM when we fit the spectra for the Ly α damping wing absorptions, there are still some possible systematics that we could consider for more accurate estimation. For example, the results from semi-numerical simulations suggest that our method assuming homogeneous IGM could overestimate the x_{HI} values by $\sim 30\%$ at the middle stage of cosmic reionization (Mesinger & Furlanetto 2008a,b). Keating et al. (2023) also suggest the simple analytical formulation using Miralda-Escudé (1998) does miss out some of the physics related Ly α absorption that are seen in

the simulation results (i.e., the contributions of residual H I gas inside the ionized bubble to the Ly α optical depth). On the otherhand, the results from different numerical simulation with various box size yield different predictions on the evolution of key cosmic reionization parameters (i.e., Keating et al. 2023; Lu et al. 2023), indicating better interpretation of the Ly α damping wing absorption measurements requires further improvements in both simulation and observations.

Another source of systematics is a biased sampling of galaxies in physical properties such as UV-luminosity. Although we could use the simulation prediction suit-

able for the observed galaxies (e.g., UV-bright galaxies residing at overdensed region; Lu et al. 2023) to infer cosmic reionization parameters, we ideally need to know the optical depths of as many unbiased sight lines as possible to precisely estimate the volume-averaged neutral fraction of the universe with a minimum dependency on the simulation models. Moreover, we must investigate the evolution and diversity of physical properties of the intrinsic UV spectrum of galaxies that are still not well studied above $z > 7$ (e.g., UV-slope, ionizing photon production efficiency, etc.) With increase in size of $z > 7$ galaxy sample, we could construct composite galaxy spectra from galaxy subsamples with more homogeneous physical properties (i.e., M_{UV}) as it has been done for $z < 6$ galaxies (e.g., Cullen et al. 2019, 2020). Analyzing a composite spectrum from a uniform galaxy sample has a potential for the estimation of average Ly α IGM absorption with high S/N at the same time minimizing the impact of variations in intrinsic spectral shapes.. In addition to the uncertainty in UV spectrum of galaxies, uncertainty in Ly α lines are also crucial in interpretation of the damping wing features seen in low-resolution PRISM spectra. Although most of the $z > 7$ Ly α detected galaxies have the velocity offset similar to that for our model (i.e., 200 km/s; Nakane et al. 2023), there are some reports of $z > 7$ galaxies with large Ly α velocity offsets as large as ~ 500 km/s (e.g., Tang et al. 2023). Such a large Ly α velocity offset could result in higher Ly α transmission, which then mimic the spectra of galaxy surrounded by large ionized bubble in low-resolution data (e.g., Keating et al. 2023). We conduct same analysis as described in Section 3.1 but with Ly α velocity offset fixed at 500 km/s, and confirmed that x_{HI} and $\log R_b$ estimates for velocity offset at 200 km/s and 500 km/s are consistent within errorbars for all redshift bins. The bubble size estimate for the two lower redshift bins (i.e., redshift bins with $\log R_b/\text{cMpc} \sim 2$ for fixed Ly α velocity offset at 200 km/s) with the velocity offset at 500 km/s change only by $\lesssim 0.2$ in $\log R_b$ values, while errorbars remain as large as those for 200 km/s. This reflects the fact that we do not marginalize the intrinsic Ly α equivalent width over any other physical values, thus the effect of change in velocity offset is compensated by a change in the intrinsic Ly α emission strength. This demonstration suggests the importance of sophisticated Ly α emission modeling based on both observation and simulation for precise damping wing parameter measurements.

In terms of the systematics from data reduction, the damping wing analysis with low-resolution PRISM data introduces uncertainties in spectroscopic redshift and the degenerating components around Lyman break (e.g.,

Ly α emission, Ly α absorption by IGM/CGM). Additionally, as Jones et al. (2023) shown, NIRSpec/PRISM spectra have systematics in the wavelength resolution depending on the compactness of the object. This could also be a potential source of the systematics in the redshift determination and damping wing absorption measurement using PRISM spectra. The redshift uncertainty of $\Delta z \sim \pm 0.01$ as suggested by Jones et al. (2023) give a slight systematics of ~ 0.1 for HPDI of x_{HI} for the individual galaxy. This systematics is insignificant compared to the current large uncertainty of x_{HI} (~ 0.5).

Lastly, we investigate how the instrumental broadening in PRISM spectrum affect Ly α damping wing profile fitting using real observation data. Here, we consider a spectroscopically confirmed galaxy at $z \sim 9$ called Gz9p3 (e.g., Boyett et al. 2023) as a test case. Gz9p3 is a suitable object to check how the instrumental broadening affect the parameter estimation for damping wing absorption, because its UV-continuum and Lyman break are detected in the high spectral resolution ($R \sim 2700$) data (Boyett et al. 2023; Harikane et al. 2023). JWST/NIRSpec data of Gz9p3 is obtained in the observation of GLASS (ERS-1324; PI. T. Treu; Treu et al. 2022). Gz9p3's spectral data is taken in high dispersion grating filter pairs of F100LP/G140H, F170LP/G235H, and F290LP/G395H, with exposure time of 4.9 hours for each configuration. In this paper, we use the high dispersion grating data reduced by Harikane et al. (2023). We adopt the spectroscopic redshift of $z = 9.313$ determined using [O II] $\lambda 3727$ and [Ne III] $\lambda 3869$ (Harikane et al. 2023). We use 1D spectrum data in the wavelength range of the 1.00-1.37, 1.70-1.95, 2.10-2.47, and 2.90-4.19 μm . To generate the mock PRISM spectrum of Gz9p3, we forward model by conducting 1000 steps of Monte Carlo simulation. For each step of Monte Carlo simulation, we fluctuate the original high dispersion 1D spectrum according to the error spectrum and then convolve to the resolution of PRISM accordingly using the PRISM line spread function provided by STScI. For both high dispersion grating and low dispersion prism data, we conduct the spectral fitting in the same manner as described in Section 3.3. We show observed and corresponding fitted model spectrum for high dispersion grating and PRISM data for in the top and bottom panel of Figure 12, respectively. In Figure 13, we show the posterior PDF of the parameter related to the Ly α damping wing absorption for Gz9p3. The results for high dispersion grating and PRISM data are shown in the black and red, respectively. As shown in Figure 13, 1D/2D marginalized PDF for high dispersion grating and PRISM data are consistent within 68% percentiles. Both high-dispersion grating and PRISM

atics to narrow down the range of possible cosmic reionization scenarios.

3. We compare estimated x_{HI} and R_b values with the theoretical predictions of the typical ionized bubble radii at different x_{HI} . Our R_b estimates lie up to a few dex above the cosmic average values estimated from analytic calculations by [Furlanetto & Oh \(2005\)](#), while our R_b estimates are comparable with the values for overlapping ionized bubbles around bright galaxies predicted by numerical simulations of ([Lu et al. 2023](#)).

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APPENDIX

A. POSTERIOR PROBABILITY DISTRIBUTION FUNCTIONS INCLUDING NUISANCE PARAMETERS

To briefly highlight the importance of considering the intrinsic galaxy's property on precise inference of globally defined parameters (e.g., $x_{\text{H I}}$, $\log R_b$), we show the posterior PDFs of global IGM parameters together with the parameters related to the intrinsic Ly α strength and the host galaxy's H I absorption strength in Figure 14. We derive the posterior PDFs for $\log EW_{\alpha,0}$ and $\log N_{\text{H I}}$ in the same manner as for $x_{\text{H I}}$ and $\log R_b$. We note, however, that $\log EW_{\alpha,0}$ and $\log N_{\text{H I}}$ are nuisance parameters and unique to the individual galaxies. Therefore, we only refer binned posterior PDFs for $\log EW_{\alpha,0}$ and $\log N_{\text{H I}}$ in terms of how the nuisance parameters could affect the inference of $x_{\text{H I}}$ and $\log R_b$. As an example, the marginalized 2-D posterior PDF for $\log R_b$ and $\log EW_{\alpha,0}$ for $z = 7.44$ shows clear correlation between the two parameters. This correlation can be explained by how strong Ly α emission could mimic the high transmission (i.e., large ionized bubble), which leads to the degeneracy between the two parameters (e.g., Keating et al. 2023) and the large uncertainty.

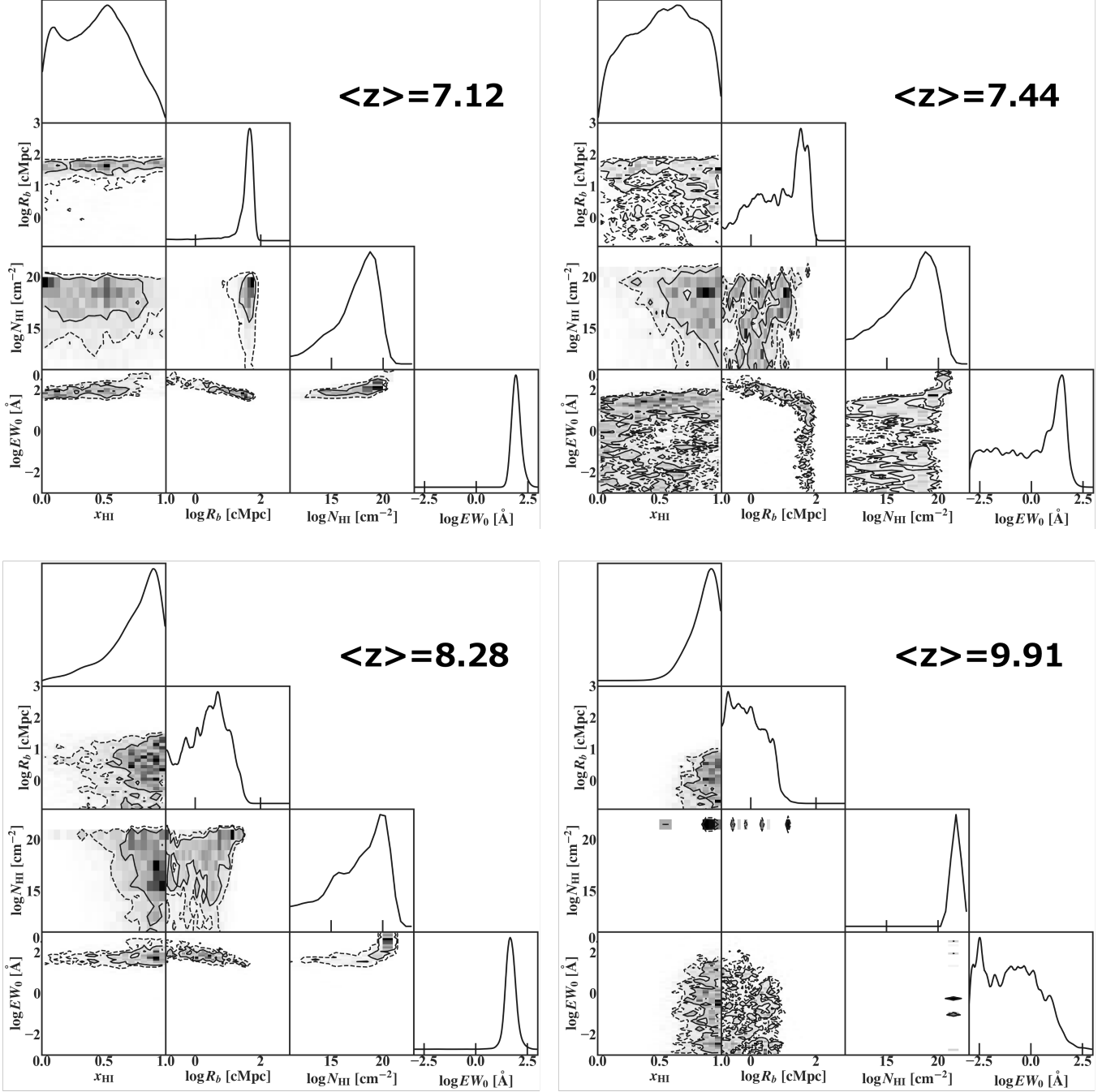


Figure 14. The posterior probability distribution function (PDF) of Ly α emission and Ly α absorptions by H I gas in the CGM/IGM for each subsample. The results for #1 to #4 bin are shown in order of top left to right bottom. The darkness in the 2D-marginalized PDFs represent the probability density. The solid and dotted contours represent the 68 and 90-th percentiles, respectively. Top panel corresponds to the 1D marginalized PDF for each parameter. The presented PDFs are smoothed by gaussian kernel.